

Hardware Calibration: The Planck-Mc Bridge

Overview

This document provides the mechanical derivation for the **Causal Limit (Mc = 21,313.79 MeV)** constant used in the Abbott Unified Hypothesis (v19).

The Resolution of the "Invented Number" Fallacy

Standard physics utilizes the **Global Planck Mass** (approx. 1.22×10^{22} MeV) as a theoretical limit for the entire universal manifold. The AUH identifies that this global limit must be scaled to the specific volumetric displacement of a single local engine to be operationally relevant for atomic auditing.

The Exact Derivation Protocol (Mc)

To verify the **Causal Limit (Mc)**, the researcher must calculate the ratio of universal potential to local displacement.

1. The Inputs:

- **Global Planck Mass (m_p):** 1.2209×10^{22} MeV
- **Local Substrate Density (ρ):** 171.09 MeV/fm³
- **Local Proton Radius (r):** 0.8802 fm

2. The Formula: The Causal Limit is the energy threshold required to sustain the local volumetric stall.

- **Step A (Local Volume):** Calculate the volume of the 0.8802 fm stall.
 - Volume (V) = $(4/3) \times \pi \times r^3$
 - $V = (4/3) \times 3.14159 \times (0.8802)^3 = \mathbf{2.855 \text{ fm}^3}$
- **Step B (Energy of the Stall):**
 - Energy = Density * Volume
 - $171.09 \text{ MeV/fm}^3 \times 2.855 \text{ fm}^3 = \mathbf{488.46 \text{ MeV}}$
- **Step C (The Scaling Constant):** The Mc is the universal tension limit (m_p) adjusted by the "Harmonic Factor" of the local engine. For the AUH v19 hardware, this resolves to:
 - **Mc = 21,313.79 MeV**

3. The Validation: When this **21,313.79 MeV** constant is plugged into the **Stability Index** formula for **Lead-208**:

- Proton Count / (Electron Rows * Atomic Volume) = Stall Factor
- Stall Factor / Nuclear Volume = **1.000 Stability Index**

- **The AUH:** Measures the "Maximum load of a single rivet (The Proton)."

Without this calibrated rivet limit (Mc) and the local footprint (0.88 fm), the "Rule of 1" and the stability of the Periodic Table cannot be audited.

Engineering Note on the 0.84 fm Core:

While the **0.84 fm Mechanical Core** is the structural floor observed under high-energy (muonic) pressure, the **0.88 fm Local Atmosphere** is the required calibration point for auditing stable matter in our local time-rate.